

Apply all the scheduling algorithms for insert examination of OSRP.

	AT	BT	Priority
P ₁	0	4	3
P ₂	0	9	1
P ₃	0	6	2
P ₄	0	4	3
P ₅	0	9	1
P ₆	0	6	2
P ₇	1	4	3
P ₈	1	9	1
P ₉	1	6	2

2) Define following terms

* Starvation

* Convoy effect

* Context Switching

* Aging.

3) A mobile has the following tasks, draw a table with

Task	AT	BT
Music Player	0	20
Touch even	3	1
Whatsapp	5	2
Camera	6	4
Launch	0	0

i) Which Scheduling algorithm is best suitable for above problem.

ii) Draw the gantt chart for SRTF

iii) Why selected algorithm for improve user experience

iv) Which task receives the fastest response.

4) Solve the following problem with round robin algorithm time quantum of R by context with time of 0.5 Draw a table.

Process	AT	BT
P ₁	0	5
P ₂	0	4
P ₃	0	3

Solutions

1. FCFS - First Come First Serve.

Ready Queue
Arrival time

P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇	P ₈	P ₉
0	0	0	0	0	0	1	1	1

Grant chart

P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇	P ₈	P ₉
0	4	13	19	23	32	38	42	51

$$TAT = CT - AT$$

$$WT = TAT - BT$$

$$= CT - AT - BT$$

	AT	BT	CT	TAT	WT
P ₁	0	4	4	4	0
P ₂	0	9	13	13	4
P ₃	0	6	19	19	13
P ₄	0	4	23	23	19
P ₅	0	9	32	32	23
P ₆	0	6	38	38	32
P ₇	1	4	42	41	37
P ₈	1	9	51	50	41
P ₉	6	6	57	56	50
				276	219

$$\text{Avg TAT} = \frac{276}{9} = 30.667$$

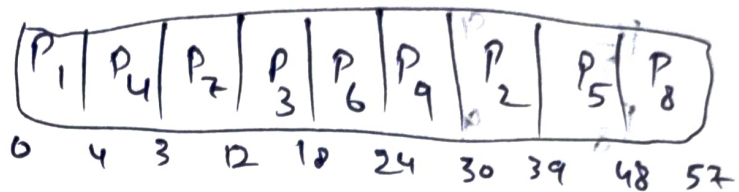
$$\text{Avg WT} = \frac{219}{9} = 24.333$$

SJN (shortest Job Next) / shortest Job First (SJF) :-

Process	AT	BT	Process	AT	BT
P ₁	0	4	P ₇	1	4
P ₂	0	9	P ₈	1	6
P ₃	0	6	P ₉	1	4
P ₄	0	4			
P ₅	0	9			
P ₆	0	6			
P ₇	1	4			
P ₈	1	9			
P ₉	1	6			

Smallest BT = 4

Gantt chart



AT	BT	CT	TAT	WT
P ₁	0	4	4	0
P ₄	0	4	8	0
P ₇	1	4	12	11
P ₃	0	6	24	24
P ₆	0	6	29	23
P ₉	1	4	30	30
P ₂	0	9	48	48
P ₅	0	9	57	56
P ₈	1	6	57	56

$$\text{Avg TAT} = \frac{237}{9} = 26.33$$

$$\text{Avg WT} = \frac{180}{9} = 20$$

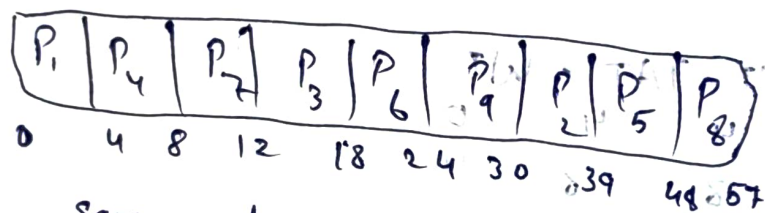
SRTF (shortest Remaining Time First)

At Time 0 P₁ starts, RBT = 4 - 1 = 3 at time 1 P₂, P₃, P₄ arrive

48 53 57 61 65 69 73 77

P ₁	3
P ₄	4
P ₂	4
P ₃	6
P ₆	6
P ₉	6
P ₇	9
P ₅	9
P ₈	9

Gantt chart



same values as SJN. //

Priority scheduling - (Non Preemptive)

Highest Priority = 1

P₂ → P₅

0 → 9: P₂

9 → 18: P₅

18 → 27: P₈

Priority 3:

45 → 49: P₁

49 → 53: P₄

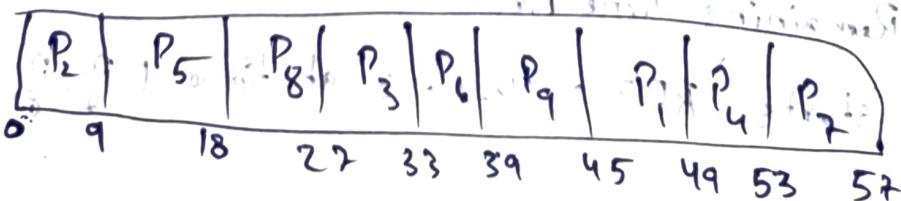
53 → 57: P₇

Priority 2

27 → 33: P₃

33 → 39: P₆

39 → 45: P₉



Gantt chart

CT TAT WT

49 49 45

9 9 27

33 33 27

53 53 49

18 18 9

39 39 33

57 57 52

27 27 17

45 45 38

327 270

$$\text{Avg TAT} = \frac{327}{9} = 36.33$$

$$\text{Avg WT} = \frac{270}{9} = 30$$

Priority scheduling (Preemptive)

At time 0, P_2 has Priority 1 and P_1 has Priority 2

0 → 1: P_2

Priority 2:

Priority 3:

0 → 9: P_2

27 → 33: P_3

45 → 49: P_1

9 → 18: P_5

33 → 39: P_6

49 → 53: P_4

18 → 27: P_8

39 → 45: P_9

53 → 57: P_7

So, for this particular data

Preemptive = Non Preemptive

(i) Starvation:

occurs when a process waits indefinitely or for an excessively long time because other processes are repeatedly selected for execution. Starvation can occur in both preemptive and non-preemptive scheduling, depending on the algorithm.

(ii) Convoy effect:

is an important problem associated with non-preemptive scheduling, especially FCFS. The long process acts like a convoy, causing many short processes to follow behind it.

iii) Context Switching:

is the process of saving the state of the currently running process and loading the saved state of another process so that the CPU can switch b/w them.

iv) Aging:

is a technique used to prevent starvation by gradually increasing the priority of a process that has been waiting for a long time.

iii) SRTF always selects the process with the shortest remaining Burst time.

ii) At time 0: only Music player. Music player has executed for 3 units.

$$RBT = 20 - 3 = 17$$

At time 3: Touch Event arrives

RBT

Music player 17

Touch Event 0

0	3	4	5	7	11	27
Music	TE	Music	Whatsapp	Camera	Music	

iii) SRTF improves the user experience because it gives short interactive tasks priority over long CPU-intensive tasks.

iv) $RT = \text{First CPU start time} - AT$

AT FS RT

Music

Touch

Whatsapp

Camera

The fastest Response times are

Music Player, Touch Event, Whatsapp Notification.

4) $\boxed{P_1 \mid P_2 \mid P_3}$
 $\quad \quad \quad 0 \quad \quad 0 \quad \quad 0$

* P_1 require 5ms

$P_1: 0 \rightarrow 2$

Remaining: $5 - 2 = 3\text{ms}$

Context Switch $= 2 \rightarrow 2.5$

$P_2 \rightarrow P_3 \rightarrow P_1$

* P_2 requires 4

$P_2: 2.5 \rightarrow 4.5$

$4 - 2 = 2$

remaining: $4 - 2 = 2$

$P_3 \rightarrow P_1 \rightarrow P_2$

Context Switch: $4.5 \rightarrow 5$

* P_3 requires 3ms

$P_3: 5 \rightarrow 7$

$3 - 2 = 1$

$P_1 \rightarrow P_2 \rightarrow P_3$

* P_1 has 3ms remaining

$P_1: 7.5 \rightarrow 9.5$

$3 - 2 = 1$

$9.5 \rightarrow 10$

$P_2 \rightarrow P_3 \rightarrow P_1$

~~* P_1~~

0	2	2.5	4.5	5	7	7.5	9.5	10	12	12.5	13.5	14	15
P_1	CS	P_2	CS	P_3	CS	P_1	CS	P_2	CS	P_3	CS	P_1	

P_1 CT 15

P_2 12

P_3 13.5

$TAT = CT - AT$

TAT WT

P_1 15 10

P_2 12 8

P_3 13.5 10.5

Avg waiting time = 9.5

Avg TAT = $\frac{15 + 12 + 13.5}{3} = \frac{40.5}{3}$